

Rhino Credit Standard and Specification (RCSS)

Version 1.1

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The Rhino Credit Standard (RCS) is a framework for measuring relative contribution in team-based work under a fixed-sum (100%) constraint.

The Rhino Credit Standard Specification (RCSS) formally specifies the rules, definitions, and computational engines required for compliance with RCS.

Fixed-point allocation itself is not new. RCSS formalizes a specific interpretation of fixed-sum allocation: relative contribution is treated as a conserved quantity expressed in percentages, and all compliant computations preserve the invariant that the team total equals exactly 100%.

RCSS is an independent measurement specification: it defines how relative contribution is represented and normalized without tying the measure to a specific tool, workflow, or grading scheme. Contribution is defined before any downstream use such as grades, money, or rewards. The same conserved measure can therefore be interpreted by different policies while preserving a consistent 100% total.

Scope, Applicability, and Responsibility

RCSS applies where multiple contributors jointly produce a coherent shared outcome that can meaningfully be treated as one whole. It should not be applied across unrelated or heterogeneous activities that do not form such a shared outcome. RCSS is a technical measurement specification. It does not prescribe organizational policy, decision-making authority, or legal compliance obligations. Responsibility for the use, interpretation, and consequences of RCSS-based implementations rests with the implementing and operating entities.

Executive Summary

Teams produce shared outcomes, but most systems lack a consistent way to determine *who contributed how much*. Existing peer-assessment and performance tools often mix grading policy with measurement, allow totals greater than 100%, or change meaning across platforms. This creates confusion for students, instructors, managers, auditors, and HR systems.

The Rhino Credit Standard (RCS) introduces a simple governing law:

All contributions within a team must sum to 100%.

This fixed-sum conservation principle treats contribution as a conserved quantity, independent of grades, money, or rewards. On top of this principle, RCS provides a clear and auditable method for allocating proportional credit among team members.

RCS establishes:

- a common contribution unit (Rhino Credit, RC)
- a structured representation of team contribution (Team Contribution Vector, TCV)
- one baseline computation engine and two optional experimental reference profiles:
 - Engine A: linear normalization (baseline RCSS method)
 - Profile B: experimental Bayesian stabilization for education
 - Profile C: experimental enterprise inference reference profile
- an implementation-ready data schema for software systems
- implementation guidance for educational and organizational pilots

The standard separates measurement from policy. RCS determines proportional contribution; institutions remain responsible for how contribution translates into grades, bonuses, authorship order, promotion, or compensation.

This document is Version 1.1 of the Rhino Credit Standard and Specification (RCSS). It is released as an open specification for potential use in:

- universities and schools evaluating team projects
- companies distributing bonuses or equity
- research groups allocating authorship credit
- software developers building peer-assessment or HR tools

RCSS v1.1 is intended as a working reference. Community feedback, independent evaluation, and future revisions are encouraged as the specification is tested beyond the author's own implementation.

1. Introduction

Team-based work is prevalent across education, management, and organizational settings. However, existing approaches to allocating credit and responsibility within teams often lack consistent normalization and transparency. Common issues include:

- Lack of transparent contribution measurement
- Inconsistent grading or evaluation practices across teams and institutions
- Susceptibility to interpersonal bias or dominance effects
- Distortions caused by free-riding or disproportionate contribution
- Limited interoperability between evaluation systems

The Rhino Credit Standard (RCS) addresses these issues by providing:

- A single consistent unit of team contribution
- A guaranteed 100% sum constraint
- Mathematically rigorous normalization of relative contribution
- Potential compatibility with education, HR, and performance management systems

RCS aims to play a role for contribution measurement similar to what NPS plays for customer sentiment: a simple, consistent way to standardize and compare relative contribution in teams.

2. Definitions

Rhino Credit (RC)

A Rhino Credit (RC) represents an individual's proportional contribution to a completed team event, expressed as part of a fixed-sum distribution.

Team Contribution Vector (TCV)

For a team consisting of $N \in \mathbb{N}$ contributors, define the Team Contribution Vector (TCV) as follows.

Let:

- $N \in \mathbb{N}$ denote the number of contributors participating in a completed team event.
- $i \in \{1, 2, \dots, N\}$ index individual contributors.

- $RC_i \in \mathbb{R}_{\geq 0}$ denote the Rhino Credit assigned to contributor i , representing that contributor's proportional share of contribution.
- $TCV = (RC_1, RC_2, \dots, RC_N)$ denote the Team Contribution Vector, which aggregates the Rhino Credits of all contributors for the event.
- $\mathbb{R}_{\geq 0}^N$ denotes the non-negative real vector space of dimension N

The Team Contribution Vector is subject to the normalization constraint:

$$\sum_{i=1}^N RC_i = 100.$$

2.3 Contribution Event (CE)

A Contribution Event (CE) is a discrete, completed team task for which individual contributions are evaluated.

2.4 Raw Rating Matrix (RRM)

For a team of N contributors, define the Raw Rating Matrix $R \in \mathbb{R}^{N \times N}$ as:

$$R = \begin{pmatrix} R_{11} & R_{12} & \cdots & R_{1N} \\ R_{21} & R_{22} & \cdots & R_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ R_{N1} & R_{N2} & \cdots & R_{NN} \end{pmatrix}$$

where R_{ij} denotes the rating assigned by contributor i to contributor j , and diagonal entries R_{ii} represent self-ratings.

3. Design Principles

RCS is governed by the following normative constraints:

1. **100% Conservation Rule:** The sum of all contributions within a team must equal 100%.
2. **Relative Contribution:** Rhino Credits measure proportional contribution, not absolute effort, time, or hours worked.
3. **Non-negativity:** Rhino Credits must be non-negative ($RC_i \geq 0$).
4. **Interoperability:** The standard is implementation-agnostic and may be integrated across diverse software systems.

5. Auditability: All computations must be reproducible from recorded inputs and declared normalization procedures.

4. Computational Engines

This section specifies the canonical computation methods for Rhino Credits (RC), as defined in Section 3.

RCSS defines Engine A as the baseline compliant computation method. Profiles B and C are optional experimental extensions for contexts with different data and governance requirements. All profiles preserve the same conserved contribution unit and the 100% normalization rule.

4.1 Engine A: Linear Normalization (Baseline Standard)

Engine A is a linear normalization engine that converts arbitrary contribution signals into a conserved contribution allocation summing to 100%, preserving relative rankings while eliminating inflation, scale distortion, and grade overshoot.

Applicability

Engine A is the baseline RCSS-compliant computation method. It is intended for lightweight implementations and environments with limited data availability.

Description

Engine A computes Rhino Credits (RC) by applying linear normalization to mean contribution ratings. It is fully deterministic and does not incorporate probabilistic or Bayesian inference.

Input rating scale (mandatory)

All entries of the Raw Rating Matrix (RRM) SHALL satisfy:

- integer values only
- closed interval 0–100
- decimals NOT permitted at rating stage

Formally:

$$R_{ij} \in \{0,1,2, \dots,100\}$$

These values represent the percentage of total team contribution that rater i believes contributor j provided.

This is not a Likert scale, rubric score, or ordinal category. It is a direct percentage claim.

100% Conservation Rule

For any valid RCSS computation:

$$\sum_{j=1}^N R C_j = 100$$

and for all contributors:

$$RC_j \geq 0$$

For each contributor j , compute the mean received rating M_j from the Raw Rating Matrix (RRM):

Step 1: Compute Mean Received Rating

For each contributor j , compute the arithmetic mean rating received:

$$M_j = \frac{1}{N} \sum_{i=1}^N R_{ij}$$

where:

- R_{ij} = percentage assigned by rater i to contributor j
- N = number of team members

Interpretation:

M_j is the central estimate of perceived contribution share before normalization.

Step 2: Linear normalization

Rhino Credits RC_j shall be computed as:

$$RC_j = \frac{M_j}{\sum_{k=1}^N M_k} \times 100$$

This guarantees that total Rhino Credits equal 100%, values remain non-negative, and results are scale-invariant under uniform inflation.

Properties of Engine A

Engine A:

- enforces the 100% conservation rule
- outputs proportional contribution shares
- prevents grade overshoot beyond 100%
- is deterministic and reproducible
- contains no Bayesian or probabilistic inference
- degrades gracefully under uniform inflation
- preserves rank order of M_j

Engine A does NOT:

- evaluate effort
- judge fairness
- apply penalties
- detect bias
- reinterpret ratings

It simply translates percentage claims to conserved contribution shares.

4.2 Experimental Profile B: Bayesian Stabilization (Education)

Profile B is an optional experimental extension of Engine A intended to explore whether Bayesian stabilization can reduce sensitivity to uncertainty, noise, and small-sample effects in educational peer-rating data.

Profile B does not replace Engine A and is not required for baseline RCSS compliance. It refines contribution signals prior to Engine A's mandatory normalization step.

This separation preserves the core RCSS conservation rule while allowing the inference method to be tested independently.

Reference requirements for Profile B

An implementation claiming Profile B compatibility shall:

- Preserve the 100% conservation rule
- Reduce sensitivity to noisy or sparse peer ratings
- Treat all contributors symmetrically in the absence of evidence
- Remain interpretable to instructors and auditors
- Converge to Engine A behavior as evidence accumulates

Any Profile B implementation that violates the 100% conservation rule is incompatible with RCSS, regardless of sophistication.

Input scale rule (no Likert)

Profile B uses the same input constraint as Engine A:

- integer values only
- 0–100 inclusive
- no decimals at the rating stage
- direct percentage contribution claims

Formally:

$$R_{ij} \in \{0, 1, 2, \dots, 100\}$$

These values represent the percentage of total team contribution that rater (i) believes contributor (j) provided.

Profile B does not allow Likert scales, ordinal rubrics, or agreement categories.

Conceptual model

Peer ratings in educational settings are typically:

- sparse
- socially biased
- unevenly calibrated
- low-signal at small sample sizes

Accordingly, Profile B treats each contributor's underlying contribution as latent; not directly observed, but inferred from percentage signals.

Let:

- R_{ij} = percentage rating given by rater i to contributor j
- $S_j = \sum_i R_{ij}$ = aggregated received percentage signal
- θ_j = latent contribution strength of contributor j

The latent variable θ_j represents the true contribution of a participant, which is not directly observable and is inferred only indirectly through peer signals.

Prior (neutral baseline)

Profile B applies an identical, weakly informative prior to all contributors:

$$\theta_j \sim \text{Gamma}(\alpha_0, \beta_0)$$

RCSS canonical educational default:

- $\alpha_0 = 1$
- $\beta_0 = 1$

Interpretation:

Before observing ratings, all contributors are assumed equal but uncertain. The prior is symmetric and minimally informative.

In Profile B, parameters α and β represent neutral Bayesian stabilizers corresponding to one virtual observation per contributor and one virtual rater, respectively. The canonical reference default ($\alpha = 1, \beta = 1$) preserves symmetry and limits overreaction to noisy peer ratings.

Alternative parameterizations may be used for domain-specific policies, but must be explicitly declared, because they alter sensitivity rather than the conservation constraint imposed by the 100% rule.

α and β are neutral Bayesian stabilizers intended to limit overreaction to noisy peer ratings while preserving proportional contribution under the 100% rule.

θ_j represents the inferred contribution strength of contributor j , estimated from peer signals under Bayesian stabilization, prior to normalization under the 100% rule.

Before observing peer ratings, all contributors are assumed equal but uncertain.

The Gamma distribution is used because it is:

- strictly non-negative
- interpretable as contribution mass
- conjugate to additive signals

- stable under aggregation

Likelihood (aggregation of peer signal)

The aggregated peer signal S_j is modeled as conditionally proportional to latent contribution strength:

$$S_j \mid \theta_j \sim \text{Poisson}(\lambda_j), \lambda_j \propto \theta_j$$

This formulation does not claim that peer ratings follow a Poisson process in reality. The Poisson–Gamma construction is used purely as a mathematical aggregation mechanism that allows noisy contribution signals to be combined and smoothed using a transparent, closed-form expression.

Posterior inference

Given conjugacy:

$$\theta_j \mid S_j \sim \text{Gamma}(\alpha_0 + S_j, \beta_0 + N)$$

The notation “ $\mid$ ” denotes conditioning on observed data; in Profile B it represents the update of contribution estimates after observing peer signals.

The posterior mean is:

$$\tilde{M}_j \equiv \mathbb{E}[\theta_j \mid S_j] = \frac{\alpha_0 + S_j}{\beta_0 + N}$$

$\tilde{M}_j$ is the Bayesian-adjusted contribution signal. This Poisson–Gamma construction is used as a minimal conjugate device to stabilize additive peer signals; no behavioral or psychological interpretation is implied.

Key properties:

- Shrinks extreme values toward equality under weak data

- Dampens single-rater dominance
- Never produces negative values
- Converges to linear aggregation as data grows

The posterior mean combines observed peer signal with a neutral baseline, producing a stabilized contribution estimate that converges to linear normalization as data increases

Mandatory normalization

Profile B shall not allocate Rhino Credits directly. Bayesian methods are used only to stabilize noisy ratings, not to decide final shares. Final allocation is always performed by Engine A so that total contribution remains exactly 100%.

All Bayesian-adjusted signals must be passed to Engine A for normalization:

$$RC_j = \frac{\tilde{M}_j}{\sum_{k=1}^N \tilde{M}_k} \times 100$$

Compliance with RCSS axioms

Normalization mathematically enforces:

$$\sum_{j=1}^N RC_j = 100$$

This conservation constraint guarantees that Rhino Credits represent a zero-sum allocation of contribution within a team, rather than an independent scoring or grading system. Bayesian inference may modify intermediate estimates, but the final normalization step mathematically enforces the invariant that total contribution remains exactly 100%.

Relationship to Engine A

As peer evidence accumulates, Profile B is intended to converge toward Engine A behavior.

$$\tilde{M}_j \rightarrow \frac{S_j}{N} \Rightarrow RC_j \rightarrow \text{Engine A result}$$

4.3 Experimental Profile C: Enterprise Reference Profile

Profile C is an optional experimental reference profile for exploring how richer inference methods might estimate relative contribution in organizational settings while preserving mandatory Engine A normalization.

Purpose and Scope

Profile C is intended for research and pilot exploration in higher-stakes organizational contexts. It has not yet been independently validated in enterprise deployment and is not part of baseline RCSS compliance. Possible contexts include:

- bonus pools
- performance-based compensation
- promotion review
- equity or profit-sharing decisions
- regulated or audited HR systems

Any Profile C implementation must preserve the core RCSS invariants, including the 100% conservation rule and mandatory Engine A normalization.

Profile C may be relevant in contexts where:

- financial consequences are material,
- auditability is mandatory,
- historical data is available,
- and governance requirements exceed those of education.

Profile C does not alter the Rhino Credit unit, the 100% conservation rule, or the normalization requirement.

Profile C does not replace Engine A or Profile B.

Instead:

- Engine A defines the baseline allocation rule
- Profile B explores single-period stabilized inference
- Profile C explores inference across raters, time, and organizational context

Profile C refines contribution estimates only. Final allocation is always enforced through Engine A normalization.

Input scale rule (reference profile; no Likert)

Profile C uses the same input form as Engine A and Profile B:

- integer values only
- 0–100 inclusive
- no decimals at rating stage
- direct percentage contribution claims

Formally:

$$R_{ij} \in \{0, 1, 2, \dots, 100\}$$

These values represent the percentage of team output rater (i) believes contributor (j) provided.

Profile C does not use Likert scales, ordinal rubrics, or agreement categories. Inputs are direct percentage estimates.

Conceptual Model

In organizational settings, peer ratings may not be independent or equally reliable.

Profile C therefore explores a Bayesian-network approach that can incorporate:

- contributor latent strength
- rater reliability
- self-assessment bias
- historical performance signals
- temporal stability

Let:

- θ_j = latent contribution strength of contributor j
- RRM = Raw Rating Matrix
- $\mathcal{B}$ = rater bias and reliability profiles
- $\mathcal{H}$ = historical contribution data

The purpose of Profile C is to explore whether these additional evidence channels can improve contribution estimates while retaining the same fixed-sum conservation rule and explicit governance constraints.

Enterprise Inference Components

A Profile C implementation may include, but is not limited to, the following modules.

Rater Reliability Weighting

Rater reliability weighting scales the influence of rating signals based on observed consistency and stability over time, without excluding any rater or altering the conservation constraint. Each rater i may be assigned a reliability weight w_i , inferred from:

- historical agreement with group consensus
- variance patterns across evaluations
- stability over time

Unreliable raters contribute less influence, without being excluded.

Self-Assessment Bias Modeling

Self-assessment ratings are treated as a distinct signal channel due to their asymmetric incentive structure and predictable bias characteristics. Profile C implementations may:

- down-weight self-ratings
- model them using explicit inflation priors
- exclude them from inference while retaining them for audit

The chosen treatment MUST be explicitly declared.

Consistency and Residual Analysis

Profile C may perform consistency and residual analysis to assess the coherence and stability of rating signals, including internal rater inconsistency, deviations from expected contribution patterns, and anomalous scoring behavior. Such analysis may inform inference confidence, rater reliability weighting, or audit flags, but must not override normalization or violate the 100% conservation rule.

Time-Series Contribution Priors

Where historical contribution data exists, Profile C may incorporate time-series contribution priors to improve stability across repeated evaluations. Such priors may reflect past contribution distributions, decay-weighted performance, and role-adjusted expectations. Temporal priors must be contributor-specific, time-stamped, and bounded to prevent lock-in or reputational entrenchment, ensuring that current evidence remains the dominant factor in inference.

Posterior Inference

Profile C computes the posterior expectation of latent contribution strength using all available evidence in the Bayesian network.

$$\mathbb{E}[\theta_j \mid \text{RRM}, \mathcal{B}, \mathcal{H}]$$

Profile C may compute a posterior estimate of each contributor's latent contribution strength from available evidence. Because this profile is experimental, all priors and assumptions should be explicit and results reproducible. The output of inference is an unconstrained contribution signal and must not be interpreted as a final allocation; Engine A normalization remains mandatory.

Mandatory Normalization (RCSS Invariant)

As with Engine A and Profile B, allocation is performed only via normalization:

$$RC_j = \frac{\mathbb{E}[\theta_j \mid \text{RRM}, \mathcal{B}, \mathcal{H}]}{\sum_{k=1}^N \mathbb{E}[\theta_k \mid \text{RRM}, \mathcal{B}, \mathcal{H}]} \times 100$$

For any RCSS-compatible allocation, final contribution shares shall be normalized to a total of 100%. Any implementation that bypasses or alters this final normalization does not preserve the RCSS conservation rule.

Auditability and Governance Requirements

Experimental Profile C implementations should support auditability by logging priors, hyperparameters, inference model versions, raw rating data, and normalization results. Where the profile is piloted in regulated or high-stakes environments, data retention, traceability, and human review requirements remain the responsibility of the implementing organization.

What Profile C Explicitly Does NOT Do

Profile C is a measurement and analytical component only. It does not, and is not intended to:

- supersede, modify, or override any organizational, institutional, contractual, or legal policy;
- determine, recommend, or enforce salary bands, compensation levels, bonuses, promotions, grades, or employment outcomes;
- replace, constrain, or automate managerial or administrative judgment;
- assign, mandate, or execute rewards, penalties, or decisions beyond the calculation of normalized contribution shares.

Profile C produces advisory contribution metrics under a 100% fixed-sum normalization framework. All interpretations, decisions, and actions based on these metrics remain the responsibility of the relevant human decision-makers and governing bodies.

Relationship among the approaches

- Engine A defines the baseline allocation mechanism
- Profile B explores stabilization under limited data
- Profile C explores richer inference under governance constraints

All three approaches:

- produce the same unit (Rhino Credit)
- obey the same conservation rule
- interoperate within the same ecosystem

Profile C Compatibility Statement

An implementation may describe itself as compatible with experimental Profile C only if it uses an explicit, auditable inference model and applies Engine A normalization as the final allocation step. This compatibility statement is not a certification.

Table 1 summarizes the baseline RCSS method and the two optional experimental profiles. Only Engine A is required for baseline RCSS compliance; Profiles B and C are reference designs intended for further testing.

Table 1:

Status	Method	Intended context	Minimum requirements
Baseline compliant	Engine A	Lightweight and educational use	100% conservation rule; non-negative RCs; valid TCV output
Experimental education	Profile B	Education with noisy or sparse ratings	Optional; explicit assumptions; reproducible stabilization; Engine A normalization
Experimental enterprise	Profile C	Organizational research and pilots	Optional and unvalidated; explicit assumptions; auditability; Engine A normalization

5. Rounding Standard

To ensure cross-system consistency and exact conservation, RCSS specifies the following rounding conventions:

- Internal precision: RC values SHOULD be stored with at least 6 decimal places.
- Display precision: RC values MAY be displayed using 1 or 2 decimal places.
- Conservation correction: If rounding results in a displayed total different from exactly 100.0, a minimal corrective adjustment SHALL be applied to one RC value to restore exact conservation.

The correction MUST preserve non-negativity and SHOULD be applied in a deterministic manner (e.g., to the largest RC value).

Example Output Format (Informative)

The following JSON structure is an illustrative example of an RCSS-compliant output. RCSS does not mandate a specific serialization format.

```
{  
  
  "teamID": "T2025-01",  
  
  "engineUsed": "B",  
  
  "members": [  
  
    { "id": "A", "rc": 38.0 },  
  
    { "id": "B", "rc": 33.8 },  
  
    { "id": "C", "rc": 28.2 }  
  
  ],  
  
  "timestamp": "2025-12-12T08:00:00Z"  
  
}
```

6. Governance and Versioning

The Rhino Credit Standard Specification (RCSS) is maintained as an open reference specification. Clarifications and version updates are currently coordinated by the author. A broader maintainer structure may be introduced if independent adoption develops.

Versioning follows these principles:

- 1.x — backward-compatible clarifications and refinements
- 2.x — extensions introducing new capabilities or optional components
- Major version changes indicate revisions that are not backward-compatible with the prior baseline specification.

Governance describes specification maintenance only and does not imply regulatory authority, certification, or enforcement.

7. Conclusion

The Rhino Credit Standard Specification (RCSS) defines a framework for representing proportional contribution within teams under a fixed-sum conservation rule. The specification formalizes terminology, a baseline normalization method, and optional experimental reference profiles so that different implementations can preserve the same underlying contribution invariant.

RCSS does not prescribe grading policies, compensation decisions, or organizational governance. It provides a measurement layer that other systems may adopt, test, or extend.

Version 1.1 is intended as a working reference. Feedback, independent evaluation, and further refinement are explicitly welcomed as the specification is tested in additional settings.